- A role, person or organisation
- Something only systems administrators can do
- Something only developers can do
- Writing Chef and Puppet scripts
- Tools

Organisations today are looking more and more towards automation to achieve faster delivery, quick feedback from customers, providing quality software, lower recovery cycle times from any crashes, minimal defects, etc. Thus, one needs to ensure that all the tools used result in better integration between the development and operations teams, helping them collaborate or communicate better.

## Industry adoption of DevOps tools

A new global study indicates that 69 per cent of enterprises in the Asia Pacific and Japan region have implemented DevOps *(https://www.ciena.com/insights/articles/Five-DevOps-habits-for-highly-successful-enterprises.html).*

Fifty per cent of organisations are now implementing DevOps. The questions and discussions with clients have shifted from, "What is DevOps?" to "How do I implement it at scale?" *(https://go.forrester.com/blogs/2018-the-year-of-enterprise-devops/).*

Facebook, Google, Twitter and Amazon are all working the DevOps way. Everything falls in place when these companies develop, test and successfully deploy code in a matter of 1 to 23,000 minutes per day (*https://devops.com/most-popular-open-source-devops-tools/*).

Also, organisations spend around 73 per cent of their DevOps budget in tool acquisition *(https://devops.com/devops-adoption-the-driving-force-of-the-industry/).*

The year 2018 has been a special journey for the IT industry, primarily because of the increasing DevOps implementation and the growing inclination towards it *(https://devops.com/integrated-tool-ecosystem-helps-implement-devops/).*

This article covers the characteristics of open source DevOps tools, and how the open source based DevOps life cycle phases are mapped to tooling when an enterprise adopts this model. It also presents the appropriate mapping of open source options with DevOps tool providers.

## Drivers for open source DevOps tools

There is no single product in the market that can fulfil the DevOps requirements across the organisation. Thus, there is a demand for a collection of tools, potentially from a variety of vendors, used in one or more stages of the life cycle.

The key reasons for DevOps tool adoption are:

- Increasing cost of ownership in the tooling landscape
- Increased complexity when integrating and managing the tools acquired to achieve DevOps
- Burgeoning legacy tools and home-grown point solutions, leading to support issues
- Tools with duplicate functionality and those that are under-utilised
- The need for a wide range of skills to support the landscape
- Process variation due to tool limitations or capability variations
- Lack of tool catalogues and usage documentation

Tool selection for DevOps implementation depends on the following design principles:

- Maximising tool usage and coverage across all services and focusing on value
- Utilising fit-for-purpose contractor tools or leveraging customer tools where required
- Integration with various sources for analytics
- Focusing on automation and productivity tools for operations
- Application performance and currency through a combination of dynamic and static analysis tools

## Characteristics of open source DevOps tools

Tools for release management, provisioning, configuration management, systems integration, monitoring, control and orchestration are important for DevOps adoption. In addition, the principles of open data, open standards and open APIs should be ingrained in the DevOps tool consolidation decision-making process. The open source DevOps tools should have the following characteristics:

- There should be no vendor lock-in and the seamless integration of enterprise wide tools, applications, products and systems developed/deployed by different organisations and vendors should be ensured.
- The DevOps tool chain should increase productivity, speed up time to market, reduce risk and increase quality.
- Vendor monopoly should be avoided, allowing for the use of free and open source DevOps tools. With data transferability across open data formats, there are greater opportunities to share data across interoperable platforms.

Adoption of open source DevOps tools enhances the interoperability with other enterprise tools because of the option of reusing recommended software stacks and libraries/components.

Figure 1 depicts the high-level characteristics of open source DevOps tools.

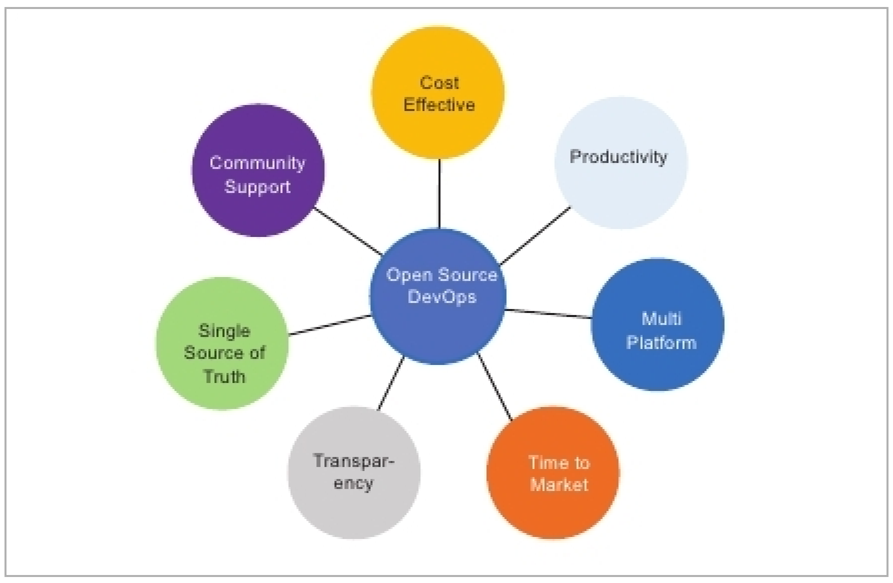


Figure 1: The characteristics of open source DevOps tools